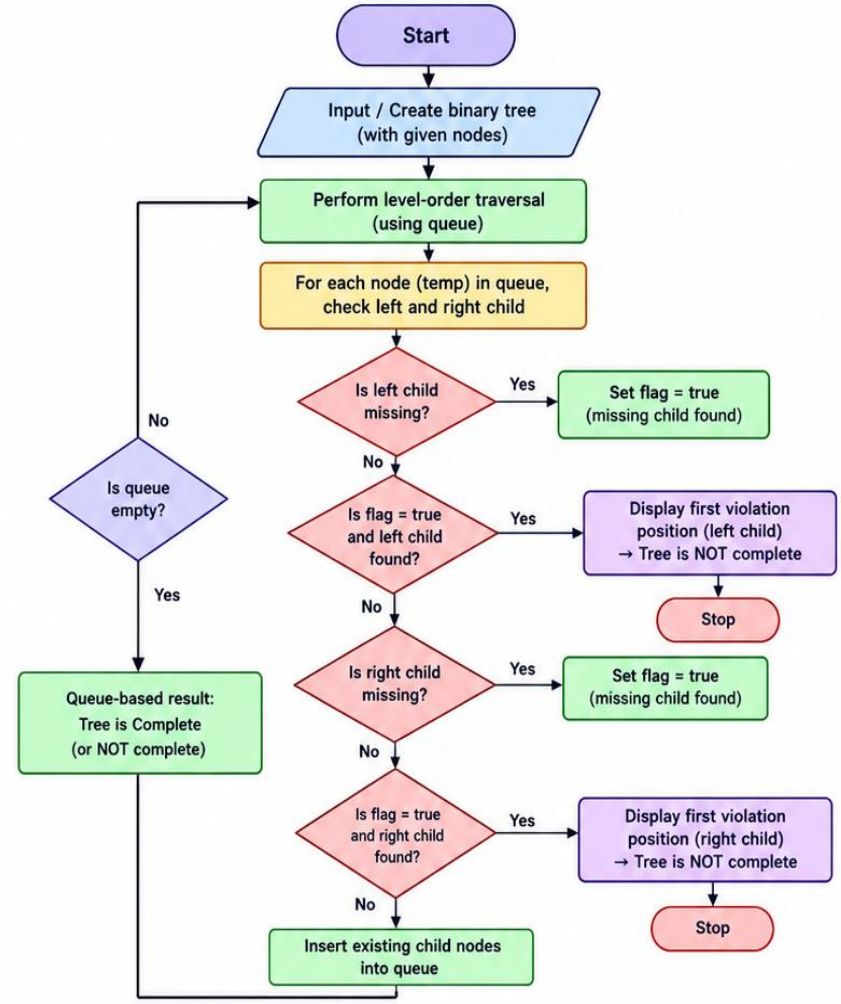


Key Points (Modifications)

1. Display the **first** position that violates completeness.
2. Test trees with missing nodes at different levels.
3. Compare the result with a completeness check using array indices.

Comparison of Methods

Feature	Queue-based method	Array-index method
Technique	Level-order traversal	Index calculation
Completeness check	Checks missing children	Checks gaps in array positions
First violation	Can report first violating child position	Can identify a gap before a later node
Time complexity	O(n)	O(n) (with indexed storage)
Space complexity	O(n)	O(n) (for array storage)

Note: In a sparse tree, array-index storage can waste space because some positions may be empty. The queue-based method is generally more space-efficient for such trees.

